



Safeguarding the Metaverse

Track:

Metaverse & Virtual World

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Subtitle:

Ensuring Safe , Respectful and Healthy Communicating Environment in the Metaverse.

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META-HARASSMENT

Virtual World, Real Pain



Problem Statement



The Metaverse is growing, but so is the toxicity which leads to the risk of harmful interactions.

Toxic Behavior

- Harassment ,verbal abuse & bullying.
- Hate speech and offensive comments.
- Trolling in voice/text chat.

Uncontrolled Actions

- Unwanted proximity
- Inappropriate gestures
- Stalking behaviors

Hidden Toxicity

- Sarcasm & slang
- Coded insults
- Non-verbal cues

Ineffective Moderation

- Slow reporting
- No real-time intervention
- Unsafe users persist

Solution/Approach Details



✗ Current Failed Solutions



Keyword Filters

- Systems only detect fixed “ bad words ” .
- They misses sarcasm, coded insults , disguised toxicity or emojis.



Reactive Reporting

- Victims must find a “**Report**” button after harassment happens.
- This offers zero protection against VR-specific issues like virtual groping or body blocking.



Cloud Processing

- 24/7 audio being sent to **cloud servers** = **huge privacy risk**, high server cost, and potential data leaks



No Gesture Detection

- Cannot recognize harmful body language.



Delayed Moderation

- Offenders stay active for too long until human moderator checks the report.

✓ Our Proposed Solution



Multimodal Sentiment Analysis

- Detects tone, sarcasm, and frustration even when actual words sound neutral.



Kinematic Proximity Shield

- Auto-creates safety bubbles, pushes offenders back and mutes them instantly.



Federated On-Device AI

- All voice and text analysis happens locally on the user’s headset/device.
- No private conversations are ever uploaded.



Gesture & Motion AI

- Detects hand signals, aggressive body movements, repeated unwanted proximity, and stalking patterns..



Real-Time Moderation Bot

- Instant action: warns , hide messages & auto mutes abusive users..

Use Case

Our solution enhances safety across diverse virtual environments.

VR Classrooms

- Prevent bullying
- Auto-mute disruptive students
- Promote safe learning environment

Gaming Platforms

- Reduce trolling and toxic voice chat
- Balanced gameplay & improved community health

Corporate VR

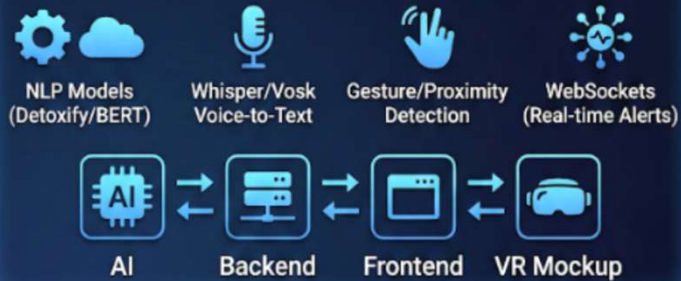
- Maintain professionalism
- Detect disrespectful language or gestures
- Improve remote work safety

Social VR

- Protect minors and women
- Reduce harassment in virtual meetups
- Increase platform trust & engagement



1. TECHNICAL FEASIBILITY



2. MARKET VIABILITY



3. LONG-TERM VIABILITY



Feasibility & Viability



Technical Feasibility

- Lightweight NLP Toxicity models.
- Speech-to-text (Whisper/Vosk).
- Raycasting for proximity detection.
- WebSockets for real-time alerts.
- Modular architecture

Market Viability

- High demand: VR classrooms, gaming ,corporate VR.
- Solves widespread online toxicity.
- SaaS safety plugin models.

Long-Term Viability

- Expand multimodal moderation
- Behavior Trust score
- Scalable across industries

Our AI moderation solution is built on robust technology, designed to meet critical market needs, and ensures sustainable growth.

Tech Stack

AI / Machine Learning

- Detoxify / BERT-based Toxicity Classifier
- Whisper / Vosk (Speech-to-Text)
- Simple Gesture & Proximity Logic

Backend

- Node.js + Express (Core API)
- Python (FastAPI / Flask)
- WebSockets (Socket.io)

Frontend

- React.js
- Three.js (Optional VR Mock Demo)

Database

- MongoDB/Firebase

Tools & Infrastructure

- GitHub
- Figma / Canva



STEP 1

✓ Handles
text + voice
moderation.

✓ Processes
messages,
runs AI
checks,
triggers
auto-
actions.

STEP 2



STEP 3

✓ Shows
real-time
flags,
warnings &
moderation
results.



✓ Stores
behaviour
logs,
flagged
messages,
trust
scores.

STEP 4



References



Google Jigsaw (Perspective API)

The Benchmark:

Provides the industry-standard dataset and toxicity scoring algorithms used to validate our detection logic.



OpenAI Whisper (2022)

Robust ASR:

Enables high-accuracy speech-to-text transcription, capturing commands even in noisy gaming environments.



Research: "Harassment in VR" (2019)

Academic Basis: The key study documenting "embodied harassment" (groping, intrusion), proving the need for spatial safety tools.



Meta Reality Labs Guidelines

Safety Framework:

Official industry standards we utilized to formally define "boundary violations" and safe interaction zones.